Reverse Engineering

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Reverse Engineering

- Reverse Engineering (RE) is a process of obtaining a geometric CAD model from 3-D points acquired by scanning/digitizing existing parts/products.

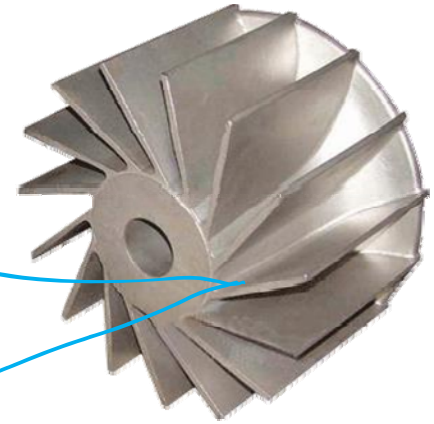
Source: https://theconversation.com/explainer-what-is-3d-printing-and-whats-it-for-9456

Reverse Engineering → ∴ Product life cycle time.

Importance:

- We cannot start from the very beginning to develop a new product every time.
- We need to optimize the resources available in our hands and reduce the production time keeping in view the customers' requirements.
- For such cases, RE is an efficient approach to significantly reduce the product development cycle.

For example: Impeller Pump Design



Source: http://www.energyinvestmentcasting.com/aerospace-4438255.html

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Is Reverse Engineering legal?

- It is also often lawful to reverse-engineer a product or process as long as it is obtained legitimately.
- If the product is patented, it doesn't necessarily need to be reverse-engineered, as patents require a public disclosure of invention.
- It should be mentioned that, just because a piece of product is patented, that does not mean the entire thing is patented; there may be parts that remain undisclosed.
- The fundamental use of Reverse Engineering is to get the feel of the product, in terms of dimensional accuracy. This helps in improvisation and to determine the flaws in the product.

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Is Reverse Engineering legal?

Justifying RE:

- The fundamental use of Reverse Engineering is to get the feel of the product, in terms of dimensional accuracy. This helps in improvisation and to determine the flaws in the product.

- In other words, the reverse engineering process in itself is not concerned with creating a copy or changing the artifact in some way; it is only an analysis in order to deduce design features from products with little or no additional knowledge about the procedures involved in their original production.

- Even when the product reverse engineered is that of a competitor, the goal may not be to copy them, but to perform competitor analysis.

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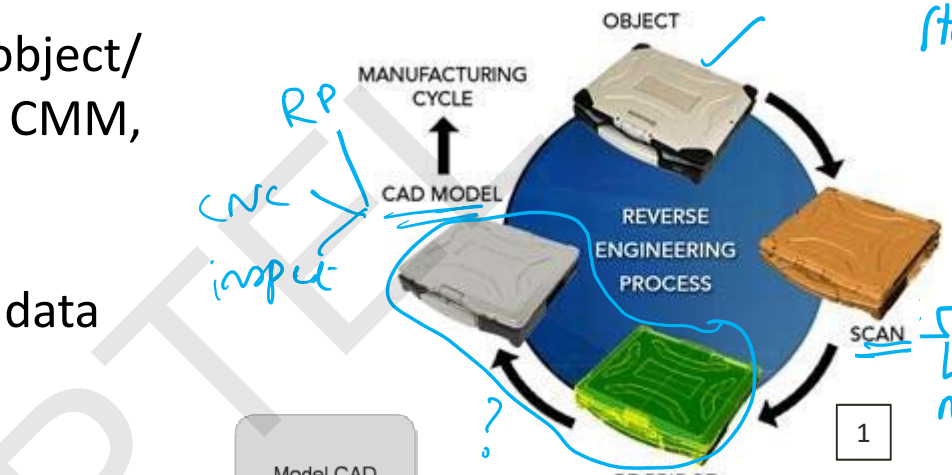
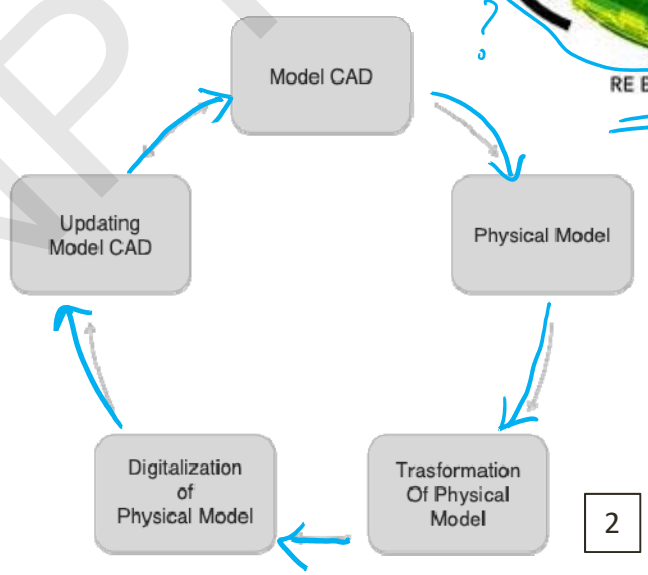
Reverse Engineering Process

1. Digitization of the object/ Data Capturing (using CMM, scanners etc.)
2. Processing of measured data
3. Creation of CAD model.
4. Prototype.



1



2

1. Source: http://www.udaipurcadatc.com/reverse-engineering/
2. Source: www.indiacadworks.com/mechanical/reverse-engineering.php

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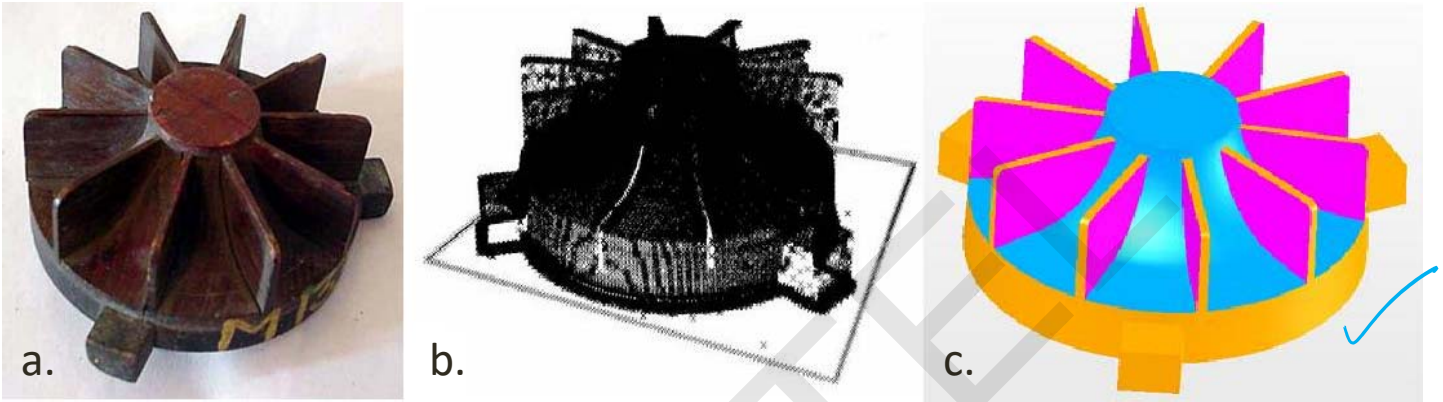
Reverse Engineering Process



CAD model generation using laser scanner:
(a) wooden pattern, (b) cloud of points, and (c) 3D CAD model

Fabricated Model

Source: http://www.energyinvestmentcasting.com/aerospace-4438255.html

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Scanning

(X, Y, Z)
Co-ordinate measuring m/c

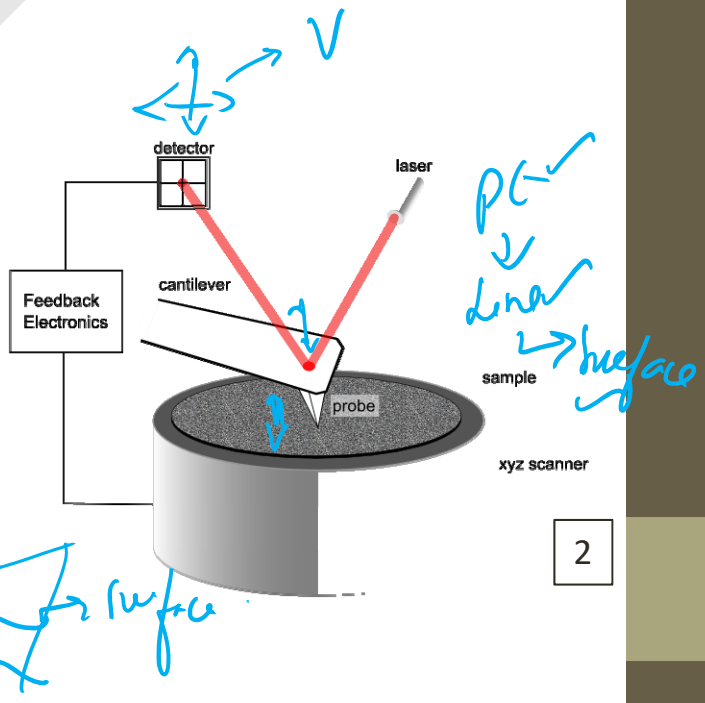
Contact Scanners:

- CMM Based.
- Soft materials can't be scanned accurately.
- Comparatively slow process.

Noncontact Scanners:

- Uses lasers, optics and CCD sensors to capture data points.
- Shiny surfaces and surfaces parallel to light axis can't be scanned accurately.

1



2

1. Source: www.cnc-step.co.za/accessories-touchprobe/
2. Source: http://afmhelp.com/index.php?option=com_content&view=article&id=51&Itemid=57

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3D Scanning Process

For an ideal scanning process, the scanning procedure has been divided into five key steps:

1. Acquisition
2. Alignment
3. Mesh Generation
4. Post Processing
5. Simplification

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3D Scanning Process

1. Acquisition

- The acquisition is the first fundamental step in which the acquired image is created in the software as a set of points.
- These points define a 3D representation of the part of the object that has been framed and hit by the light pattern generated by the projector.
- For this reason, it is advised to proceed with the acquisition of a wide part of the object first, postponing the acquisition of details and missing parts in a following moment.
- Once a rough 3D reconstruction has been obtained, the scan can be improved by adding more views that correspond to some missing parts.

Source: https://www.creaform3d.com/en/news/creaform-presents-perfect-3d-scanning--solution-3d-printing-applications

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3D Scanning Process

2. Alignment

- Alignment is the work phase, where, it is possible to bring to the same reference system (align) the Range Images (i.e. images acquired by using the scanner at different lengths from the object) acquired previously.

- Manual Alignment: The process is manually helped by the identification of three corresponding points between the two acquisitions taken into account.

- Global Alignment: Beside the manual alignment, that works with the identification of three corresponding points, another alignment tool called 'global alignment' is also available. It is advised to run this command after having manually aligned all the range images, in this way the alignment of each acquisition is optimized with respect to the others

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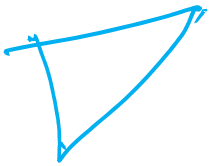
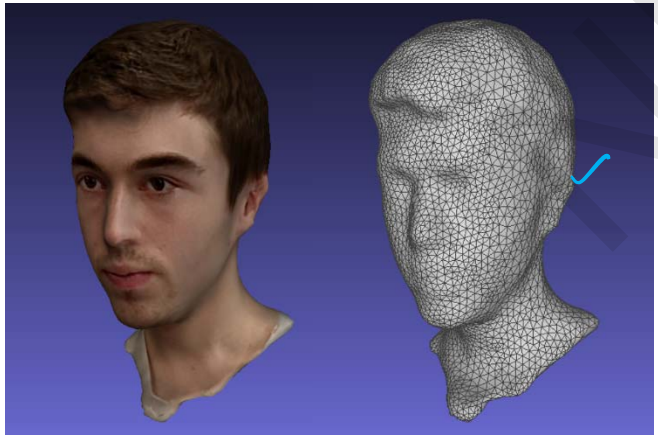
3D Scanning Process

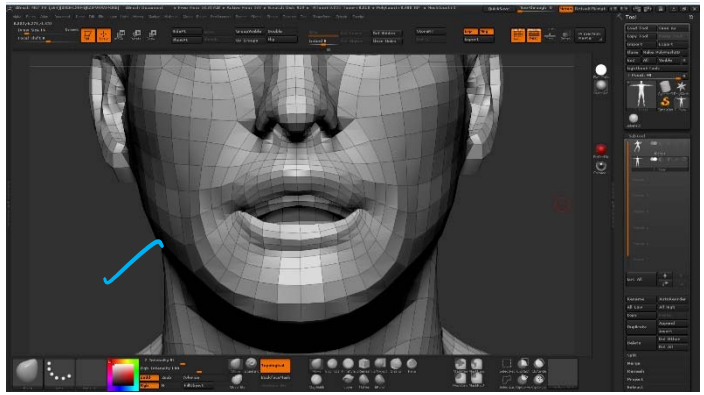


3. Mesh Generation

Once a sufficient number of range images has been acquired and aligned in order to create a 3D model as complete as possible, the next step is to generate a triangular Mesh.

- The Mesh generation converts a set of 3D points (Range Image) to a data constituted by a set of triangles (Mesh).
- The Mesh is the first useful data that can be elaborated and exported in the available formats.





1. Source: https://www.3dscanstore.com/index.php?route=journal2%2Fblog%2Fpost&journal_blog_post_id=23
2. Source: https://medium.com/shadow/from-scan-to-avatar-the-unsuspected-hurdles-618f201e22e8

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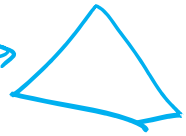
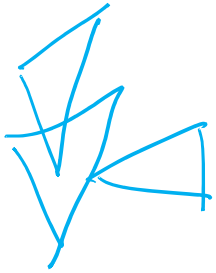
3D Scanning Process

mesh → facet → 
↓
shape edge

4. Post Processing

- Post Processing is every operation that involves the enhancement and finishing of a mesh. Its purpose is to prepare a complete and flawless 3D model ready to be exported.
- These operations should be chosen depending on the result to be achieved and they can affect more or less the 3D model.
  - Make Manifold: This involves solving the possible topological issues that can be attributed to the presence of triangle edges shared by more than two faces.
  - Detect and repair intersection: Solving some possible topological issue, attributed to the triangles that intersect other triangles of the mesh surface.
  - Fill Holes: Detection and fixing of missing parts on the mesh surface. It automatically fills the missing data with a surface composed of triangles that propagates the nearby shape and texture information.



→ Euler formula

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3D Scanning Process

CAD
- simple handle.
- store easy
- edit - easy

5. Simplification

Under this process, are gathered all the steps made on the mesh that (tend to) simplify the data.

- Reduce noise on mesh: When the generated 3D model presents some surface imperfection, such as roughness or orange peel effect, a filter commonly defined as reduce noise, can be applied to smooth the surface. This is an operation works like a digital sandpaper.
- Mesh Decimation:
  - Reduction of the number of mesh triangles. This operation can be done forcing a tolerance that guarantees that the decimated 3D model does not differ more than this value from the original model.
  - It is done in order to have a more manageable file that is quicker to elaborate with the post processing tools, and of smaller dimension, thus easier to share on the internet and with less occupancy on the hard disk, without losing the 3D model details.

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Reverse Engineering:
Hardware & Software

- RE hardware is used for RE data acquisition.
- There are three main technologies for RE data acquisition: contact, non-contact and destructive.
- Outputs of the RE data acquisition are 2-D cross-sectional images and point clouds that define the geometry of an object.
- RE software transforms the RE data produced by RE hardware into 3-D geometric models.
- RE data processing chain can be one of two types of 3-D data: (i) polygons or (ii) curves
- Polygon models are commonly used for rapid prototyping, laser milling, 3-D graphics, simulation, and animations.

- pt
- Line
- Curve

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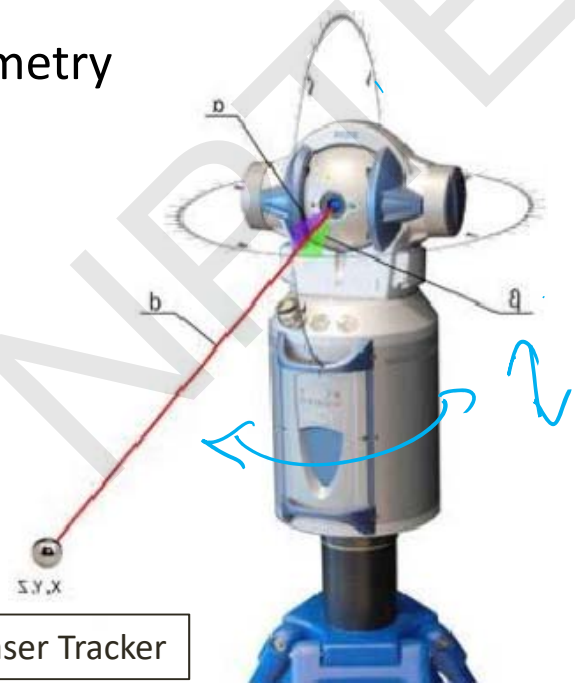
Reverse Engineering: Hardware

Classification:

- Laser Trackers
- Total Stations
- Digital Photogrammetry
- Portable CMM's

3. Total Station



2. Laser Tracker

1. Portable CMM

1. Source: http://gemini-bps.com/products_portable_cmms.html
2. Source: http://www.oasisalignment.com/blog/laser-trackers-3d-precision-measurement/
3. Source: https://tinyurl.com/ycofjr5l

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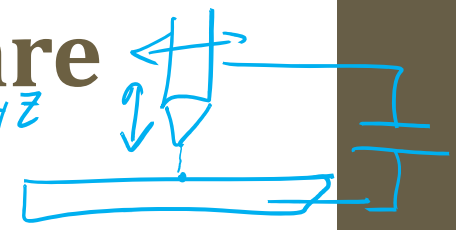
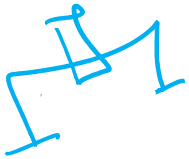
Reverse Engineering: Hardware



Contact Methods:

- Use sensing devices with mechanical arms, CMM and CNC m/c to digitize a surface.
- Data collection techniques:
  i. point to point sensing with touch-trigger probes installed on CMM or articulated mechanical arm to gather co-ordinate points of a surface.



CMM provides more accurate measurement data compared the articulate arm, but due to lack of no. of DOF, it can't be used to digitize complex surfaces.

  ii. analogue sensing with scanning probes installed on CMM or CNC m/c .

The scanning probe provides a continuous deflection output that can be combined with m/c position to derive the location of the surface.

SPM
STM
AFM

The scanning speed in analogue sensing is up to 3 times faster than point- to-point sensing.

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Reverse Engineering: Hardware

Point Processing

- Importing the point cloud data.
- Reducing the noise in the data collected. – S/N ↓ → amp/filters
- Reducing the number of points.
- These tasks are performed using a range of predefined filters.
- Multiple scans are sometimes needed to ensure all the required points are scanned.
- A wide range of commercial software are required.
- The output is a clean, merged, point cloud data set in the most convenient format.

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Reverse Engineering: Hardware

Point-to-point sensing equipment

1.

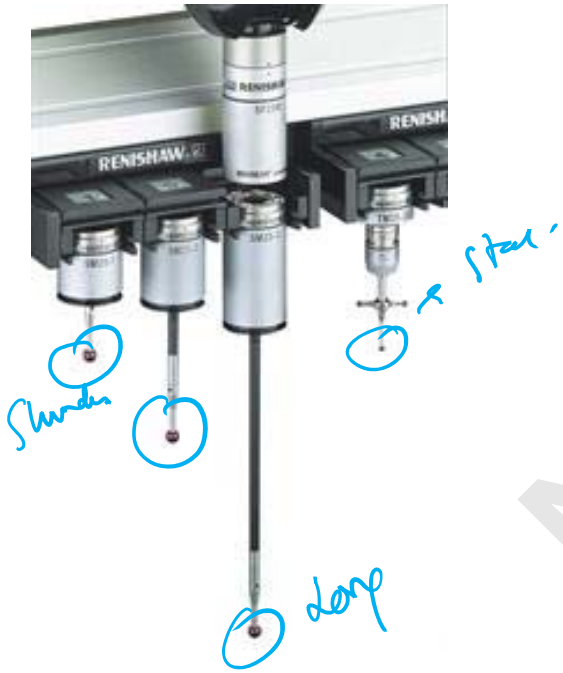
2.

Zenith 3 CNC CMM

1. Source: https://www.indiamart.com/proddetail/3d-portable-cmm-11080164633.html
2. Source: https://www.aberlink.com/products/cmm/zenith-3/

Reverse Engineering: Hardware

Analogue sensing equipments



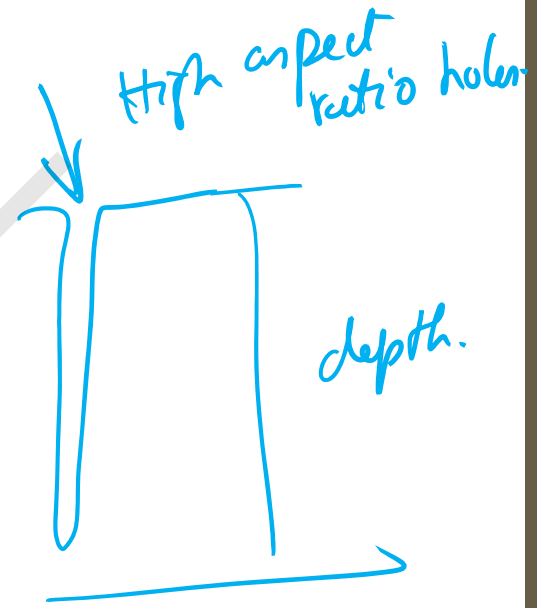
1. SP25M scanning probes from Renishaw Inc.
2. Roland DGA Corp. MDX-20 scanning and milling machine using Roland Active Piezo Sensor for 3D scanning

1. Source: https://www.renishaw.com/en/sp25m--6688
2. Source: http://www.fablab.uchile.cl/proyectos/220/tutorial-modela-mdx-20-para-fabricacion-de-pcb/

Reverse Engineering: Hardware

Advantages of contact mode

- High accuracy.
- Low costs.
- Ability to measure deep slots and pockets.
- Insensitivity to colour or transparency.



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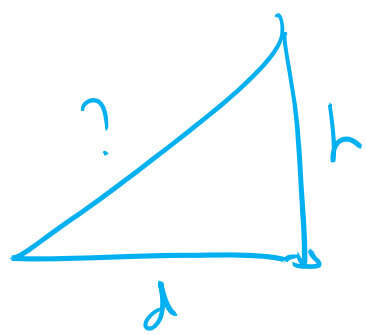
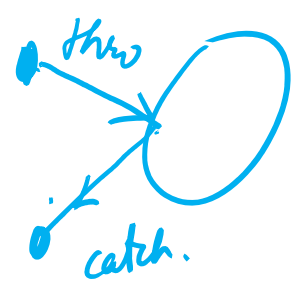
Non-contact methods

- 2D cross sectional images and point clouds are captured projecting energy sources (light, sound, or magnetic field) on an object.
- Then, either the transmitted or the reflected energy is observed.
- The geometrical data are calculated by using triangulation, time-of-flight, wave interference information, and image processing algorithms.
- The classifications of non-contact RE hardware are based on the sensor technologies or data acquisition techniques employed.





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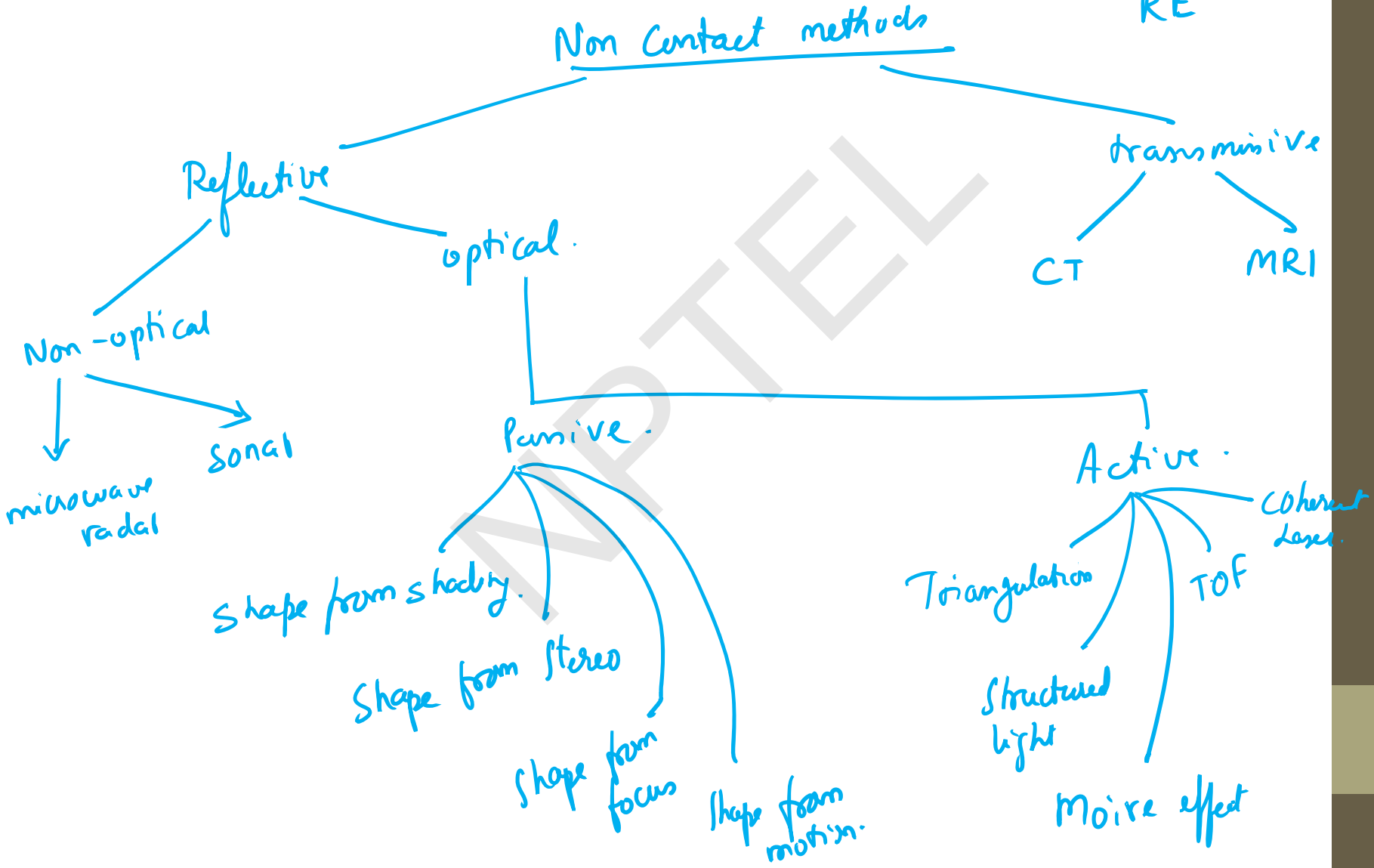
Non-contact methods –

Classification of Hardware devices used in RE



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Advantages of non-contact mode

- No physical contact.
- Fast digitizing of substantial volumes.
- Good accuracy and resolution for common applications.
- Ability to detect colours.
- Ability to scan highly detailed objects, where mechanical touch probes may be too large to accomplish the task.

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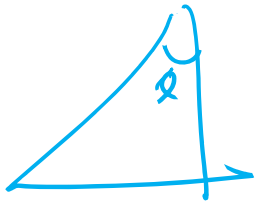
Optical techniques



Triangulation

- Most laser scanners use straightforward geometric triangulation to determine the surface coordinates of an object.
- Triangulation is a method that employs locations and angles between light sources and photosensitive devices (CCD) to calculate coordinates.
- There are two variants of triangulation schemes using CCD cameras: single and double CCD camera.
- In a single camera system, a device transmits a light spot (or line) on the object at a defined angle. A CCD camera detects the position of the reflected point (or line) on the surface.
- In a double camera system, two CCD cameras are used. The light projector is not involved in any measuring functions and may consist of a moving light spot or line, moving stripe patterns, or a static arbitrary pattern.

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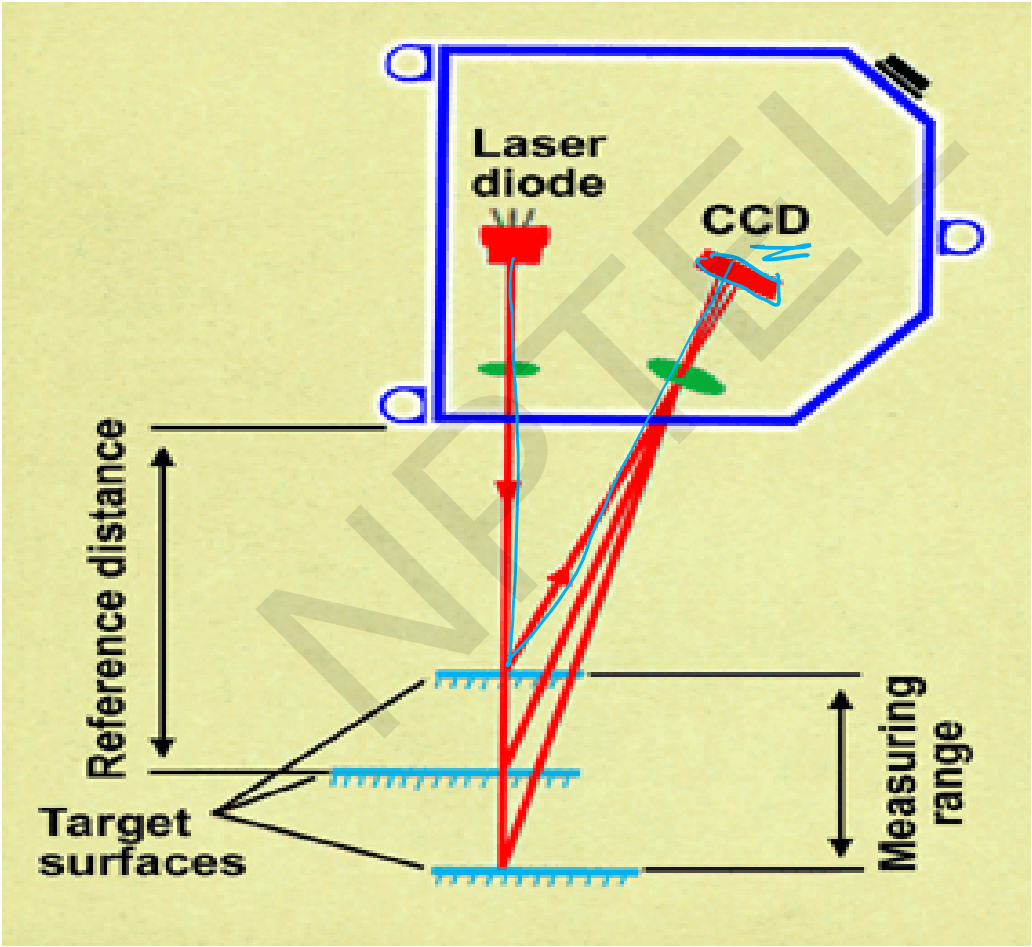
Optical Techniques

Triangulation



https://www.globalspec.com/learnmore/sensors_transducers_detectors/linear_position_sensing/optical_triangulation_position_sensors

Optical techniques

- In fig (a), a high energy light source is focused and projected at a pre-specified angle (θ) onto the surface of an object.
- A photosensitive device senses the reflection from the illuminated point on the surface.
- Fixed baseline length (L) between the light source and the camera is known from calibration.
- Using geometric triangulation from the known angle (θ), the focal length of the camera (F), the image coordinate of the illuminated point (P), and fixed baseline length (L), the position of the illuminated point (Pi) with respect to the camera coordinate system can be calculated as follows:

$$Z = \frac{FL}{P + F\tan\theta}$$

$$X = L - Z\tan\theta$$

- The measurement errors in P and θ can be determined from the following equation:

$$\Delta Z = \frac{Z^2}{FL}\Delta P + \frac{Z^2\sec^2\theta}{L}\Delta\theta$$

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Structured Light

- A light pattern is projected at a known angle onto the surface of interest and an image of the resulting pattern, reflected by the surface, is captured.
- The image is then analyzed to calculate the coordinates of the data point on the surface.
- A light pattern can be (i) a single point; (ii) a sheet of light (line) or (iii) a strip, grid, or more complex coded light.
- The most commonly used pattern is a sheet of light that is generated by fanning out a light beam.
- When a sheet of light intersects an object, a line of light is formed along the contour of the object. This line is detected and the X, Y, Z coordinates of hundreds of points along the line are simultaneously calculated by triangulation.

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Features of Structured Light

- The data acquisition is very fast (up to millions of points per second).
- Colour texture information is available.
- Structured-light systems do not use a laser.

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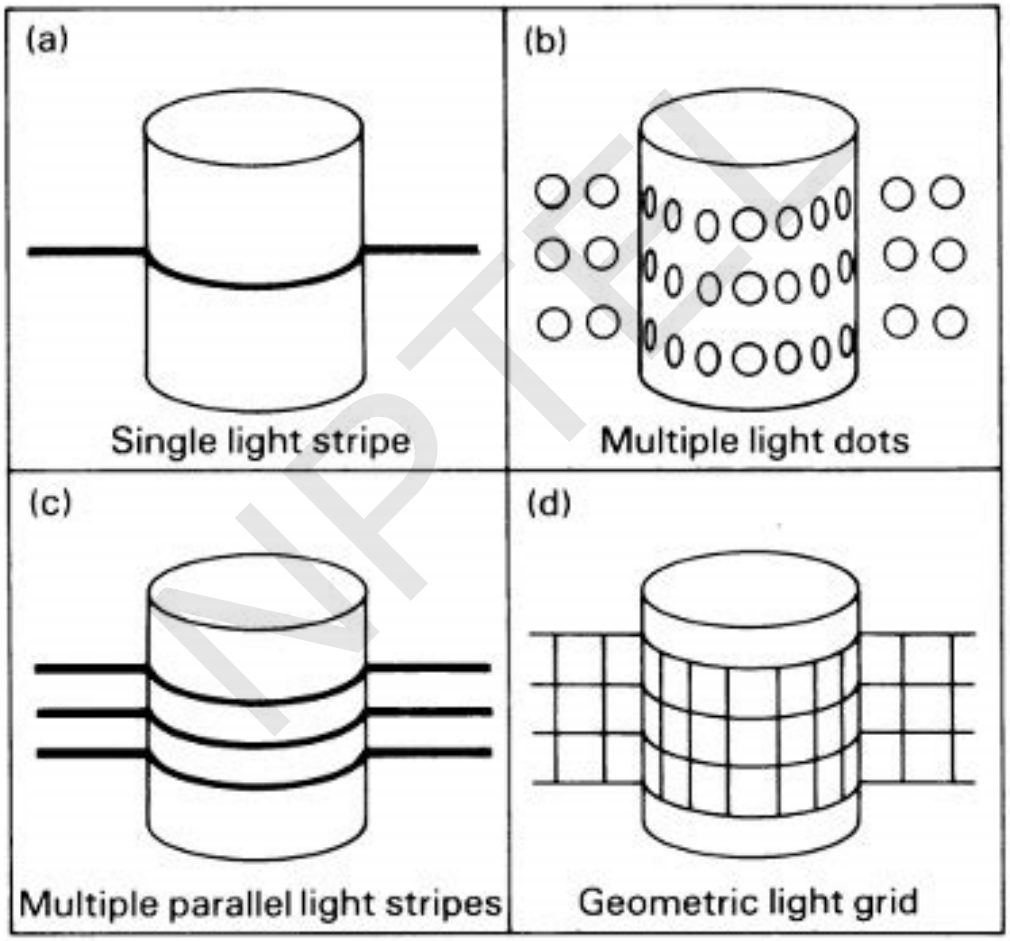
Features of Structured Light

Different light patterns used in structured-light techniques



Source: http://transducersensors.com/structured-lighting-for-3d-sensors/different-types-of-structured-light-patterns/

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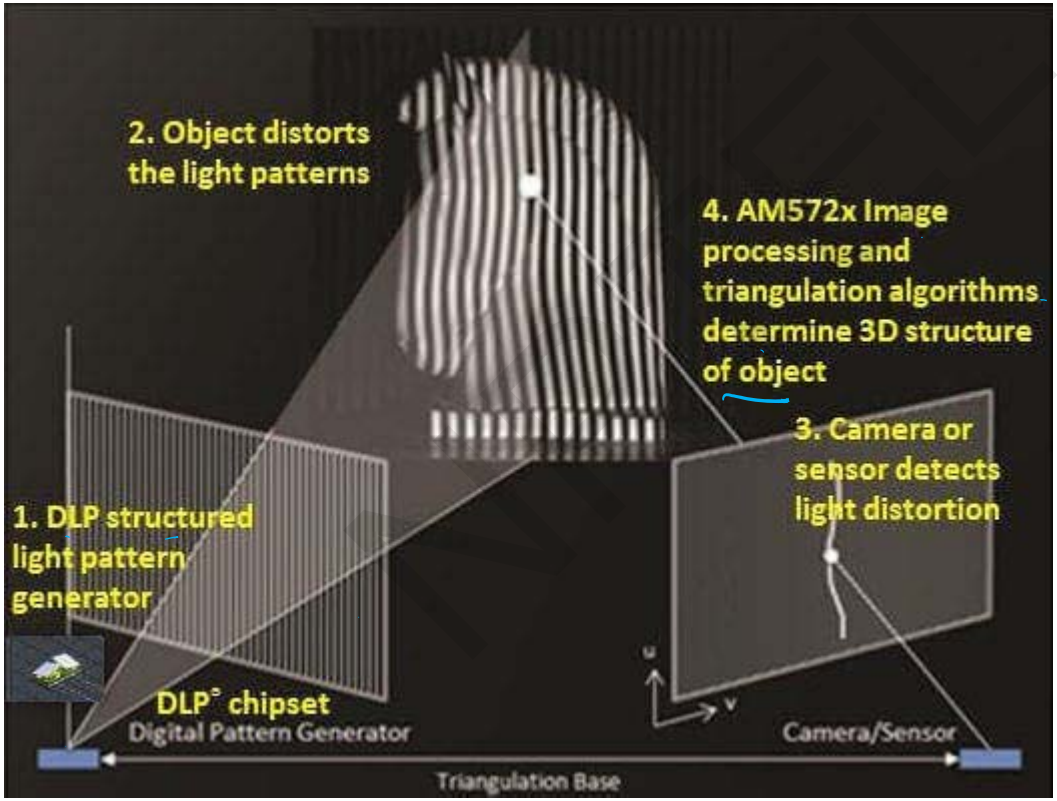
Features of Structured Light

2-D Image acquisition using structured light, parallel light strips pattern.



Source: https://all3dp.com/1/best-3d-scanner-diy-handheld-app-software

Interferometry (Moiré Effects)

- Used for dimensional inspection and flatness and deformation measurements.
- Structured-light patterns are projected onto a surface to produce shadow Moiré effects.
- The light contours produced by Moiré effects are captured in an image and analyzed to determine distances between the lines.
- This distance is proportional to the height of the surface at the point of interest, and so the surface coordinates can be calculated.
- The Moiré technique gives accurate results for 3-D reconstruction and measurement of small objects and surfaces.
- However, it has limitations for larger objects because precision is sacrificed for range.

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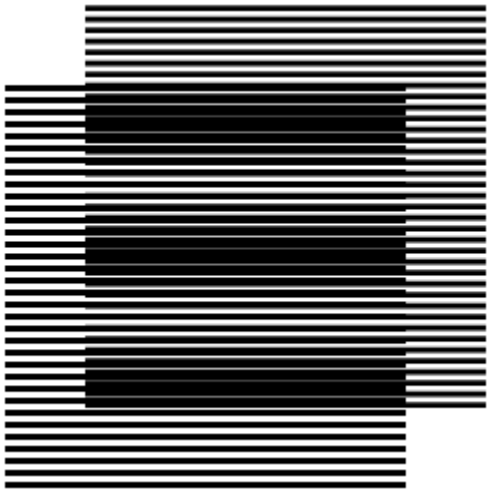
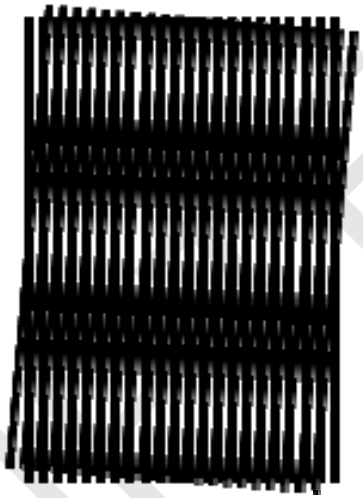
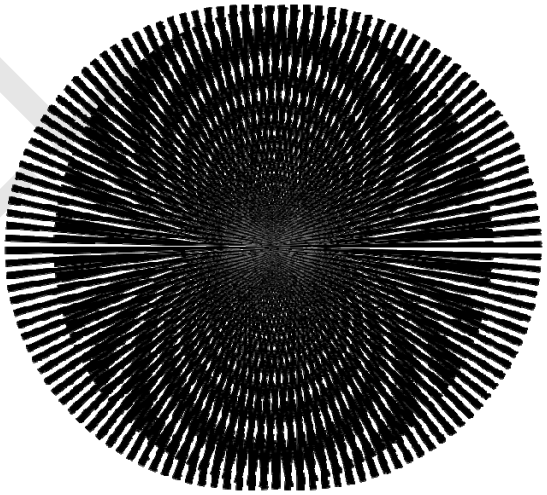
Interferometry (Moiré Effects)



1.



2. A moiré pattern, formed by two sets of parallel lines, one set inclined at an angle of 5° to the other.



3. Concentric Circles Pattern

1. Source: https://commons.wikimedia.org/wiki/File:Moir%C3%A9_pattern.svg
2. Source: https://en.wikipedia.org/wiki/Moir%C3%A9_pattern#/media/File:070309-moire-a5-a5-upward-movement.gif
3. Source: https://en.wikipedia.org/wiki/Moir%C3%A9_pattern#/media/File:Moire_Lines.svg

Destructive Method

- The RE destructive method is useful for small and complex objects, in which both internal and external features are required to be scanned.
- A CNC milling machine exposes 2-D cross-sectional (slice) images, which are then gathered by a CCD camera.
- The scanning software automatically converts the digital bitmap image to edge detected points, as the part is scanned.
- To remodel the part, 2-D slice images of the part are gathered by destroying the part layer-by-layer.

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Examples of 3D data applications

- Environmental data
  - GIS
  - Air Quality
  - Water Quality
- Built Environment
  - Building Design (CAD/Architecture modeling)
  - Building Documentation
  - Ground Penetrating Radar
- Manufactured Objects
  - Design/Production/Quality Control
  - Catalog/Sale
- Medical
  - CAT scans
  - Ultrasound

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Data Components

- Raw Data
  - Textual
  - 2D Image/surface
  - 3D Spatial
  - N-Dimensional
    - Time
    - Sensor data
- Derived Data
- Modeled Data
- Technical Data
  - File Formats
  - Applications
- Researcher Annotations (to permit reconstruction)
- “Bookmark” Information Documenting Searches/Interaction

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Metadata

- Traditional textual info (Dublin Core, COPARES, etc.)
- Non-static information (audio, video, etc.)
- 2D data (from new and existing sources)
- 3D data (point cloud and modeled)
- Multi-dimensional data (i.e. vector, sensor, time, etc.)

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Recapitulation

- What is Reverse Engineering? – Importance, Process
- What are the different types of scanners?
- What do you mean by point processing?
- How is the geometric model generated?
- What is the role of hardware and software in reverse engineering?
- What do you mean by contact method of scanning?
- What do you mean by Non-Contact mode of scanning?

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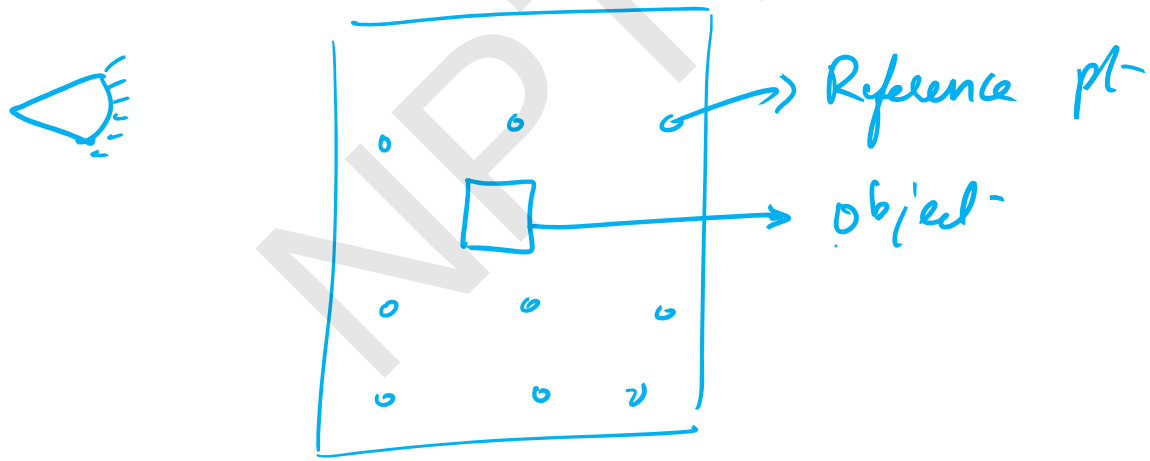
Task for Students

- Watch you-tube for "Development of 3-D Scanners"

→ Camera - Lap-top for RE



→ Try develop Urself a same technique use tennis ball as object.

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Thanks

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Managing Competitiveness

Competitiveness → relating to: characterized by or based on competition.

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Contents

- **Benchmarking**
- **Outsourcing** ⇒ reducing cost & reducing MLT
- **Mass Customisation** ⇒ batch size = 1

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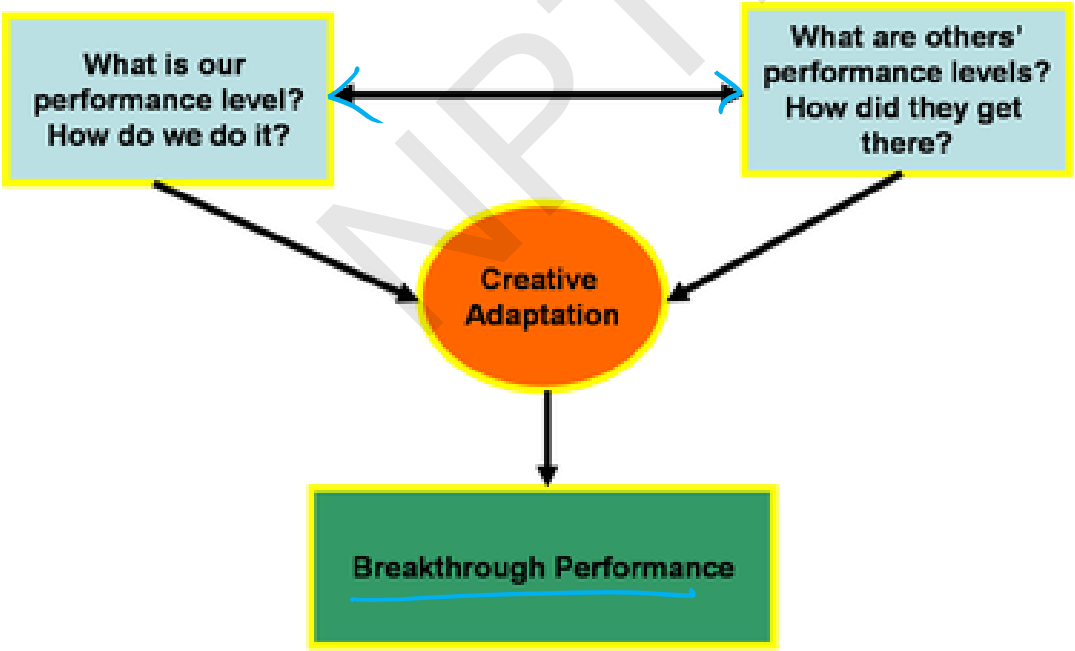
Benchmarking

- **What is benchmarking?**
- Benchmarking is a way to go backstage and watch another company's performance from the wings, where all stage tricks and hurried realignments are visible.



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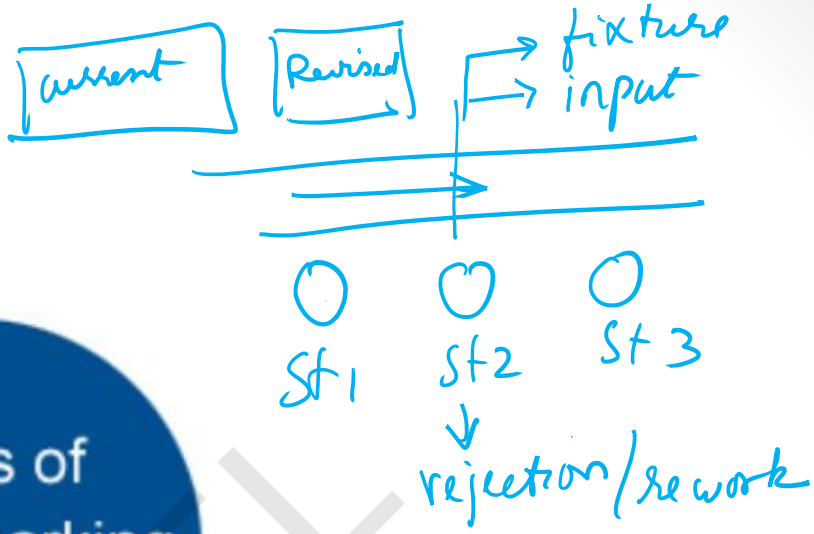
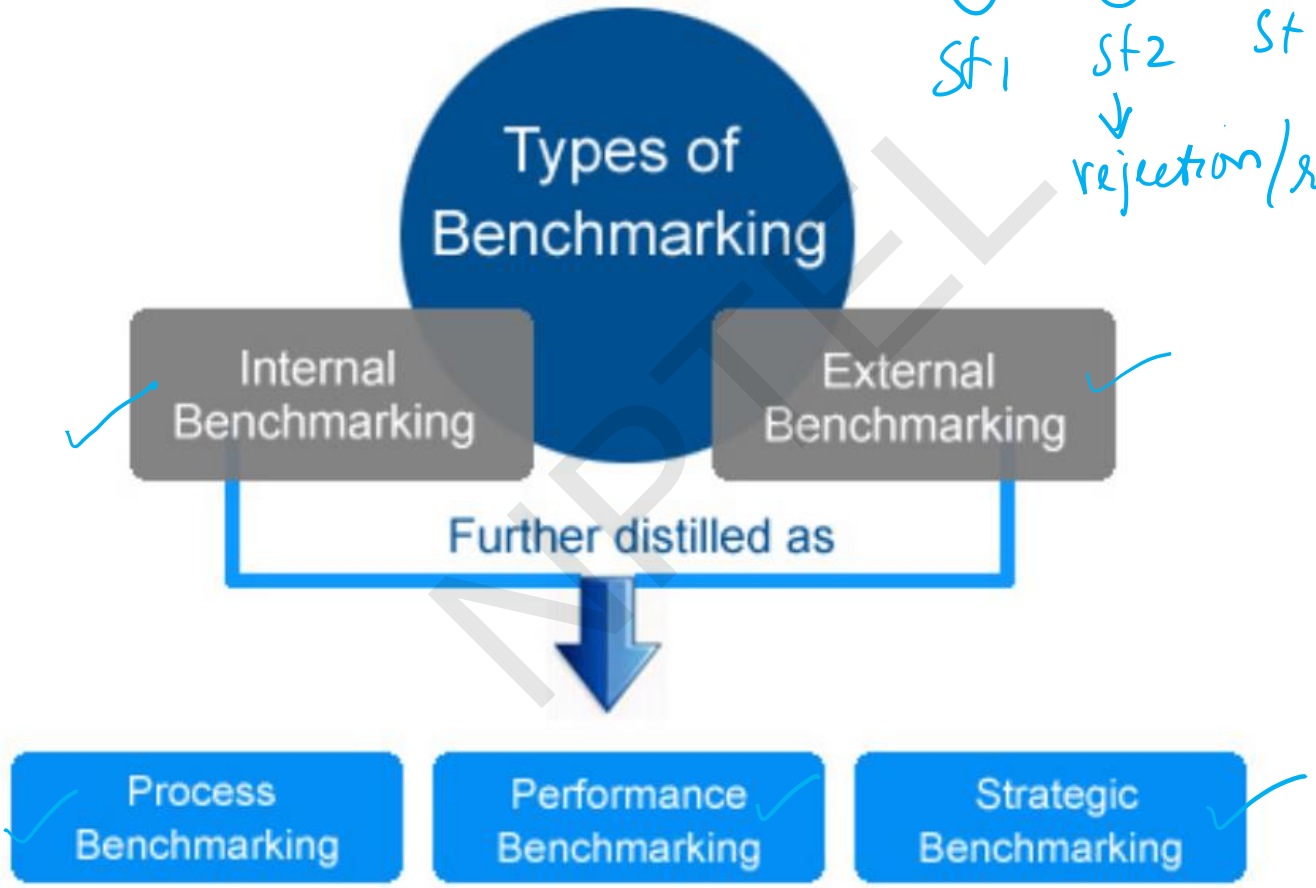
Benchmarking





http://www.best-in-class.com/bestp/domrep.nsf/insights/what-is-benchmarking-definition-types!opendocument

Benchmarking

Process Benchmarking:

- Demonstration of how top performing companies accomplish the specific process.
- Such benchmarking is gathered via research, surveys/interviews, and site visits.
- By identifying how others perform the same functional task or objective, people gain insight and ideas they may not otherwise achieve.
- Such information affirms and supports decision-making by executives.

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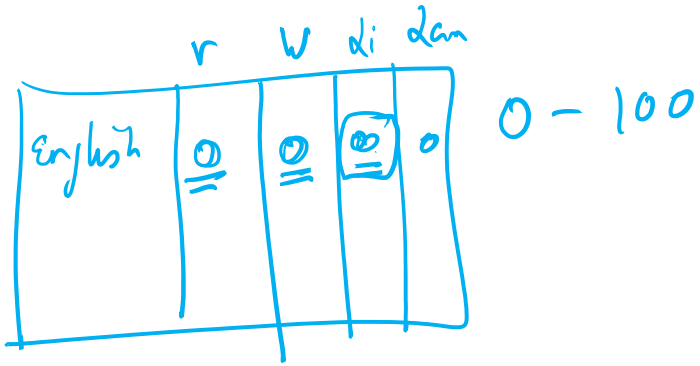
Benchmarking



Performance Metrics:

- Performance metrics give numerical standard against which a client's own processes can be compared.
- These metrics are usually determined via a detailed and carefully analyzed survey or interviews.
- Clients can then identify performance gaps, prioritize action items, and then conduct follow-on studies to determine methods of improvement.

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Benchmarking

Strategic Benchmarking:

- Identification of the fundamental lessons and winning strategies that have enabled high performing companies to be successful in their marketplaces.
- Strategic benchmarking examines how companies compete and is ideal for corporations with a long-term perspective.

nestle → maggie

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Benchmarking

The Essence of Benchmarking

- The essence of benchmarking is the continuous process of comparing a company's
- strategy,
- products,
- processes

  with those of the world leaders and best-in-class organizations.
- Main purpose is to learn how the achieved excellence, and then setting out to match and even surpass it.
- The justification lies partly in the question: Why reinvent the wheel if I can learn from someone who has already done it?
- However, Benchmarking is not a panacea that can replace all other quality efforts or management processes.

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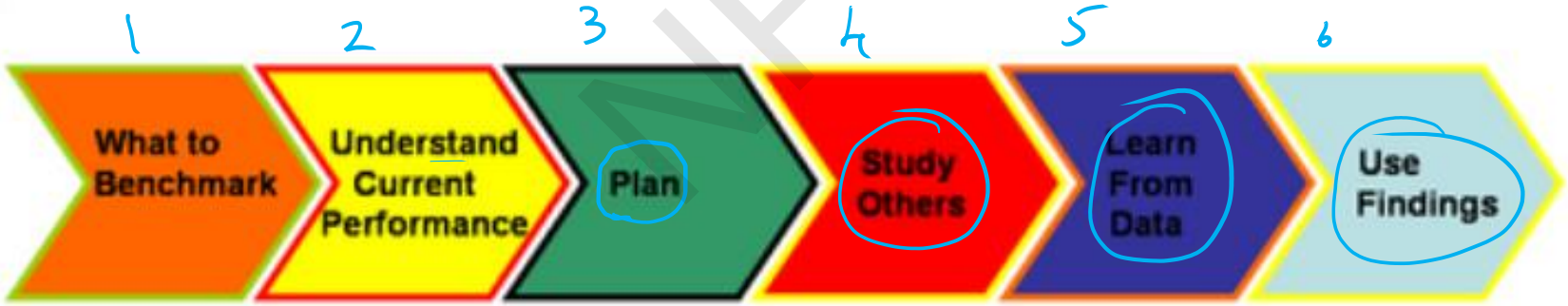
Benchmarking

Process of Benchmarking :

- Organizations that benchmark adapt the process to best fit their own needs and culture.
- Number of steps in the process may vary from organization to organization.
- The following six steps contains the core techniques.



House → Place of clothes

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Benchmarking

Levels of Benchmarking

There are three levels of benchmarking:

1. Internal benchmarking (within the company)
2. Competitive or strategic benchmarking (Industry and competitors)
3. Benchmarking outside the industry.

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Benchmarking

Benefits :

There are three sets of benefits:

1. Cultural Change:

- Benchmarking allows organizations to set realistic, rigorous new performance targets, and this process helps convince people of the credibility of these targets.

2. Performance Improvement:

- Benchmarking allows the organization to define specific gaps in performance and to select the processes to improve.
- These gaps provide objectives and action plans for improvement at all levels of organization and promote improved performance for individual and group participants.

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Benchmarking

Benefits :

3. Human Resources:

- Benchmarking provides basis for training.
- Employees begin to see gap between what they are doing and what best-in-class are doing.
- Closing the gap points out the need of personnel to be trained to learn techniques of problem solving and process improvement.

Problem solving skill / Solver

⇓

Design thinking

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Outsourcing

- In business, outsourcing is an agreement in which one company contracts-out a part of their existing internal activity to another company.
- It involves the contracting out of a business process (e.g. payroll processing, claims processing) and operational, and/or non-core functions (e.g. manufacturing, facility management, call center support) to another party.

https://newtheory.com/fantastic-2017-growth-hack-34-of-companies-are-outsourcing/

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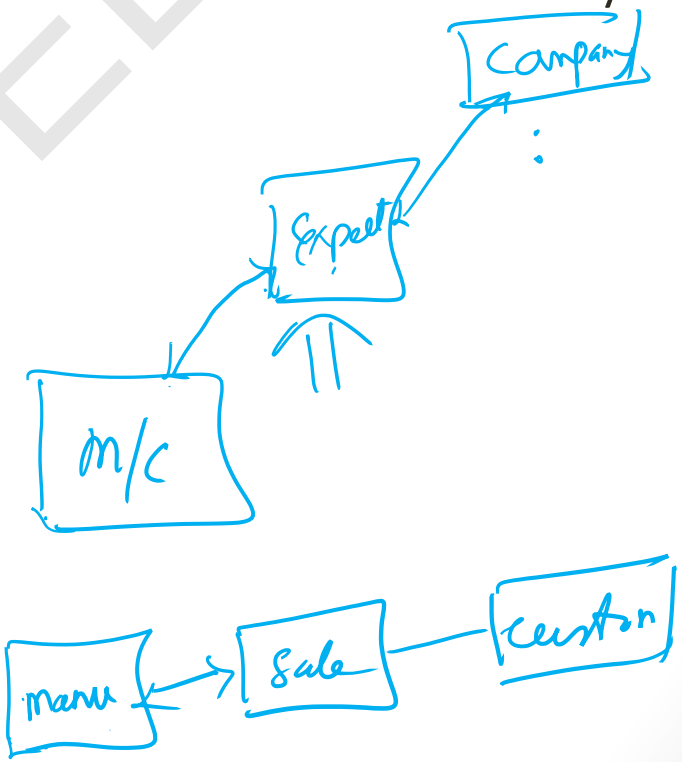
Outsourcing

Different Outsourcing Categories

- Nowadays, most freelancers or outsourcing providers will fall into one of the following categories, in regards to the services that they provide their clients.
- Some will even offer several of these:
  - Inbound Customer Service
  - Outbound Telemarketing
  - Web Design & Development
  - SEO and Online Marketing
  - Back Office / Admin Support
  - Virtual Assistant Services
  - Accounting and HR Management
  - Marketing & Sales Support



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Outsourcing

Different Outsourcing Categories

- Simply put, outsourcing will give away some of your business tasks that can easily be managed by an independent entity, making life easier for business owners.

- Most of the time, as well as being able to pass on these tasks to someone more experienced than yourself (or your company), you will also save money, against hiring someone locally to do the same job.

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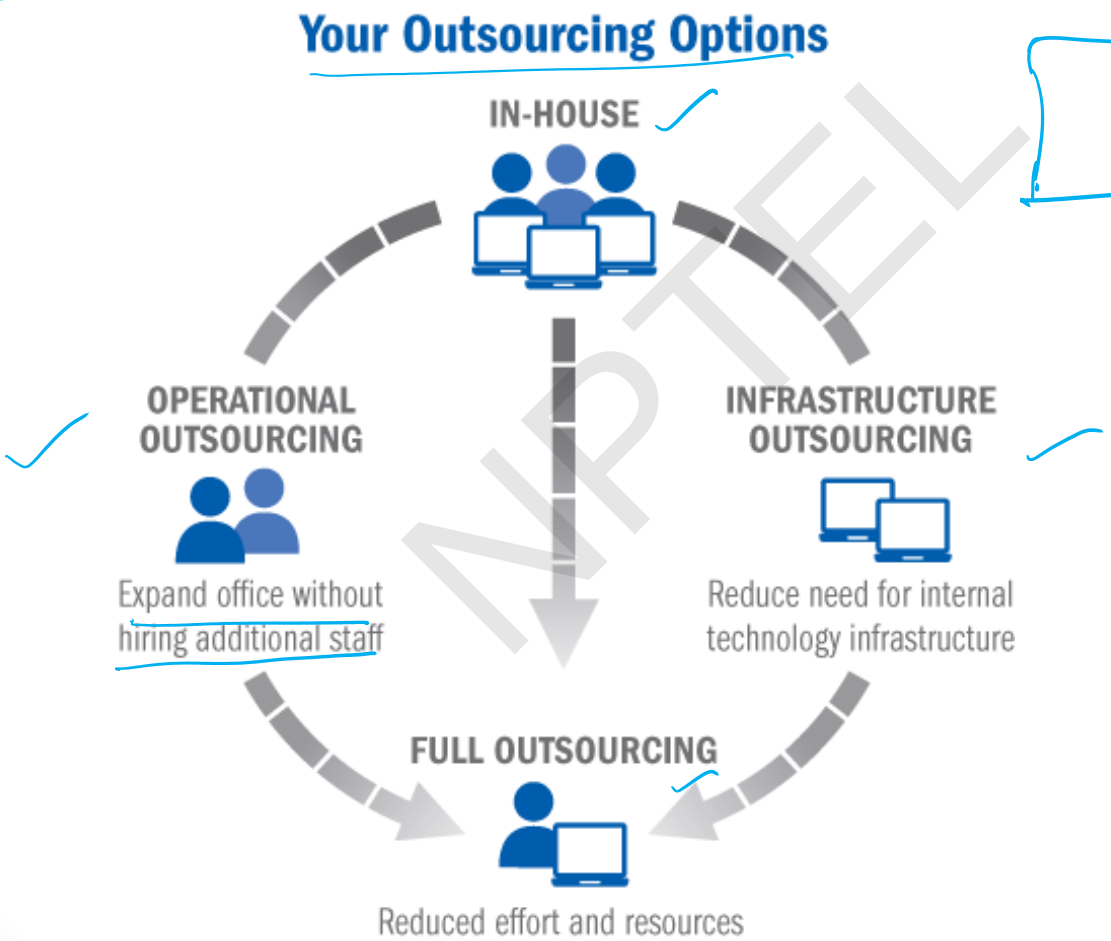
Outsourcing

Workflow



http://www.schwabintelligenttechnologies.com/OutsourcingEducation

Outsourcing

Outsourcing myths

- Outsourcing technology is expensive.
- Going to third party will mean lower quality.
- If my data is off site, I will loose control over it.

Quality

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Mass Customization

What is Mass Customization?

- Mass customization is the process of delivering wide-market goods and services that are modified to satisfy a specific customer need.
- Mass customization is a marketing and manufacturing technique that combines the flexibility and personalization of custom-made products with the low unit costs associated with mass production.
- Mass customization products may also be referred to as made to order or built to order.

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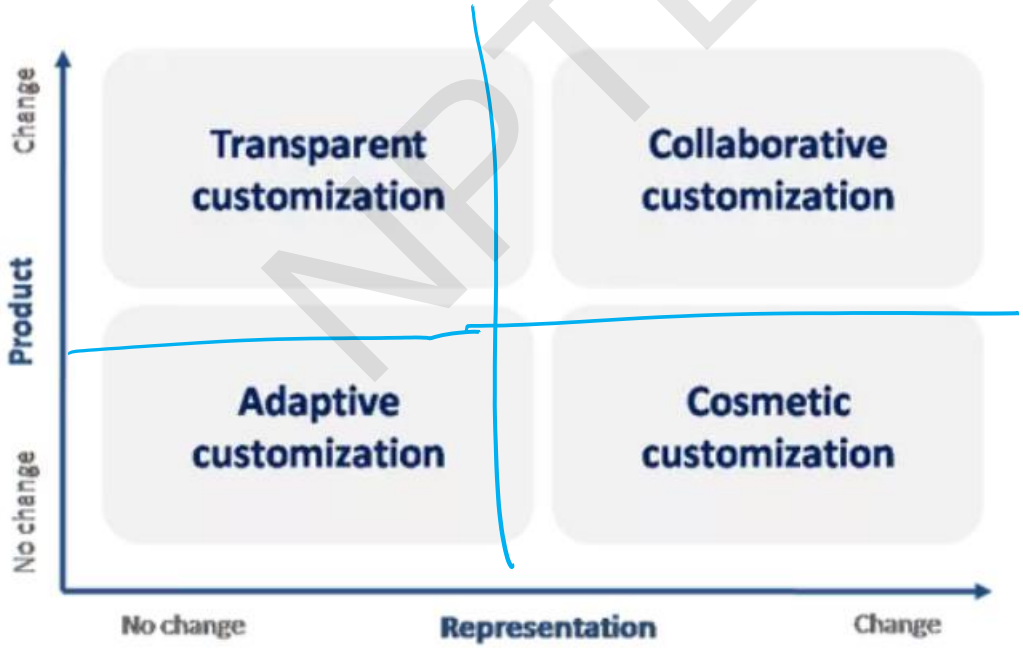
Mass Customization

Approaches to mass customization

- There are four basic approaches to mass customization, depending on customization of the product itself or its representation.



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Mass Customization

1.Collaborative Customization

- Collaborative customizers talk to the clients to help them recognize what they need, to recognize factors that will fulfill those needs and to create customized products following those guidelines.

2. Adaptive Customization

- Businesses that follow the approach of adaptive customization offer one standard product to the customers along with a few customization options.
- This approach makes sure that the product is designed in a manner that it can be customized by the end client with absolute ease.

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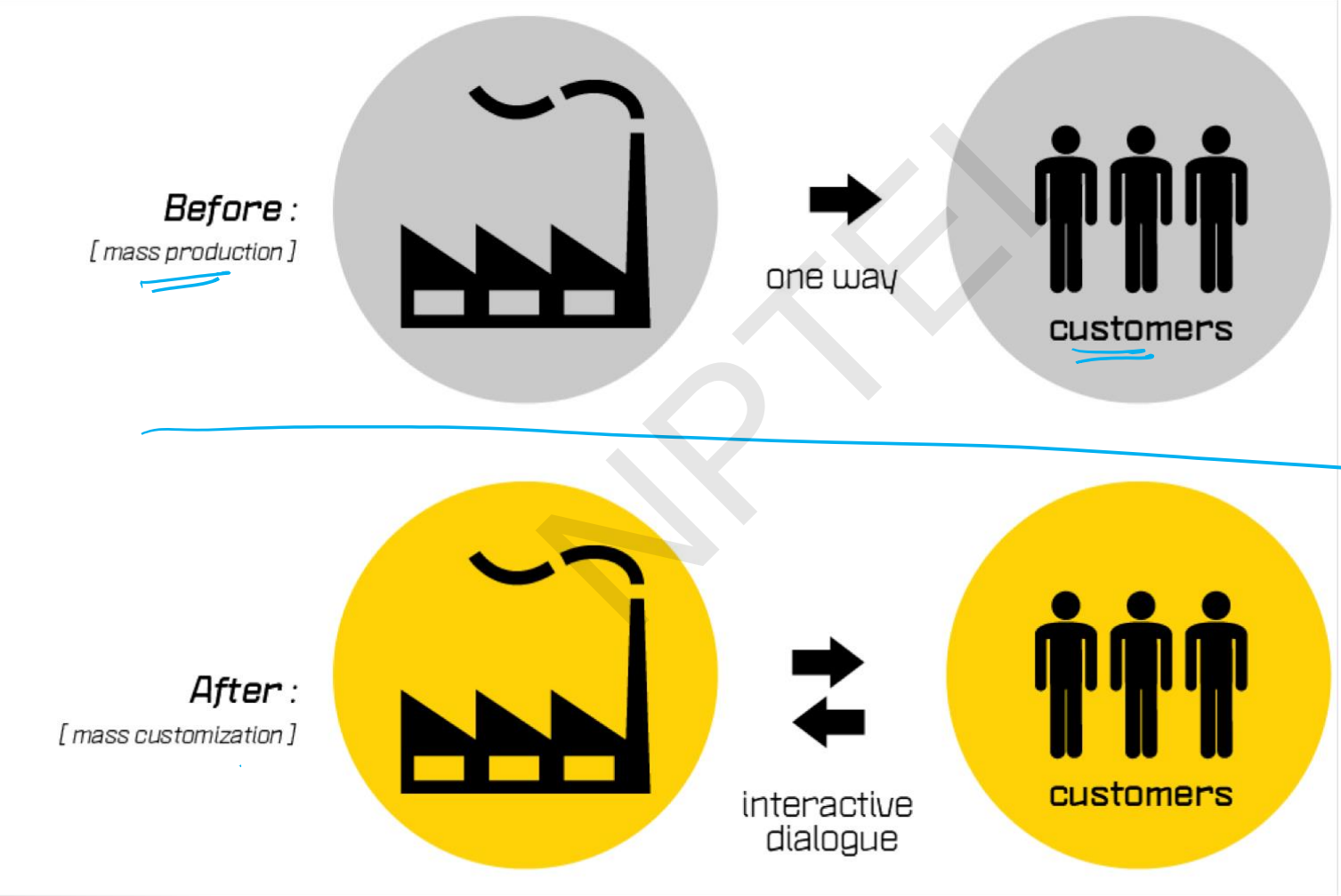
Mass Customization



Source :https://blog.marketculture.com/tag/customerizationmass-customization-business-and-consumer-interactionservice-interactionpeople-centric-businessescustomer-centric-businesses/

Mass Customization

3. Cosmetic Customization

- Cosmetic customizers advertize a standard product differently to different groups of clients.
- This approach works well when clients use the same product but want them to be presented differently.

4. Transparent Customization

- Transparent Customization deals with providing customized products to individual clients without telling them that the products are exclusively produced for them.
- This approach work in cases where the customer does not want to repeat what she/he needs incessantly or when client needs are predictable or obvious.

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Mass Customization

Advantages

1. Customer participation is more.
2. Customers are more pleased.
3. Stronger relation between company and customers.
4. Elimination of scrap quantity.

Disadvantages

1. Requirement of highly flexible production technology.
2. Difficulty in achieving efficiency.
3. Requirement of strong direct to customer logistics system.
4. High warehousing costs.

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To recapitulate:

- What is Benchmarking?
- Types of Benchmarking and its levels?
- What is outsourcing?
- Myths and reality in outsourcing
- What is Mass Customisation?
- Various approaches to Mass Customization?

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Recapitulate the PDM course

Following aspects are covered in the course:

- **Product characteristics**
- **Product Design Process, and Costing**
- **Value Engineering, Creativity, and Patenting**
- **Design for 'X'**
- **Rapid Prototyping, 3DP Laboratory demonstration**
- **Manufacturing systems, Plant Layout, software demonstration**
- **Computer Integrated Manufacturing, and Reverse Engineering**
- **Competitiveness management**

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Task for students

- Try to make a prototype of a Pencil/Pen Pouch by understanding Customer's Voice.
- You pick 1 pouch which is already avail and do modifications to meet the above requirement
- You pick 1 best product in the domain of Pen/Pencil Pouch and do benchmark (QFD)
  ↳ House of Qu

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Thanks

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